

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class :	Batch : B1
Date of Assignment : 22/7/26	Date/Time of Submission : 22/7/26

Questions:

2a. Write a python code to create nested tuples.

Code:-

```
name = ("Shweta Singh", 12, ("Bsc.IT", "Sem 3"), 90)

print("Name :", name[0])
print("Roll_no :", name[1])
print("Department :", name[2][0])
print("Semester :", name[2][1])
print("Percentage :", name[-1])

print("Shweta Singh S056")
```

Output:-

```
Name : Shweta Singh
Roll_no : 12
Department : Bsc.IT
Semester : Sem 3
Percentage : 90
Shweta Singh S056
```

2b. Write a python code to sort the nested tuple using sorted () function.

```
data = [(1, 4), (3, 2), (5, 1)]

sorted_data = sorted(data, key=lambda x: x[1])

print(sorted_data)

print("Shweta Singh S056")
```

```
[(5, 1), (3, 2), (1, 4)]
Shweta Singh S056
```

2c. Write a python code to copy or clone list.

```
list1 = (1, 2, 3, 4, 5)
list2 = 0
list2 = list1

print("list2 =", list2)

print("Shweta Singh S056")
```

```
list2 = (1, 2, 3, 4, 5)
Shweta Singh S056
```

2d. Write a python code to check immutability property of python tuples.

```
name = ("Shweta Singh", 12, ("Bsc.IT", "Sem 3"), 90)

name[0] = "Sakshi"

print("Name :", name[0])
print("Roll_no :", name[1])
print("Department :", name[2][0])
print("Semester :", name[2][1])
print("Percentage :", name[-1])

print("Shweta Singh S056")
```

ERROR!

Traceback (most recent call last):

File "<main.py>", line 3, in <module>

TypeError: 'tuple' object does not support item assignment

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80